TYPE 3: Trinomials The coefficient of x^2 is equal to 1 (a = 1).

Make notes about each of the following examples:

Example 1: $x^2 + 8x + 15 = (x + 5)(x + 3)$

Example 2: $x^2 - 8x + 15 = (x - 5)(x - 3)$

Example 3: $x^2 + 2x - 15 = (x + 5)(x - 3)$

Example 4: $x^2 - 2x - 15 = (x - 5)(x + 3)$

Exercise 5

Factorise the following completely:

1. a. $x^2 + 2x + 1$ $(x+1)^2$ b. $x^2 + 4x + 3$ $(x+3)(x+1)$ c. $x^2 + 5x + 4$ $(x+4)(x+1)$
   d. $x^2 + 3x + 2$ $(x+2)(x+1)$ e. $x^2 + 8x + 7$ $(x+7)(x+1)$ f. $x^2 + 12x + 11$ $(x+11)(x+1)$
   g. $x^2 + 20x + 19$ $(x+19)(x+1)$ h. $x^2 + 6x + 5$ $(x+5)(x+1)$

2. a. $x^2 - 7x + 6$ $(x-6)(x-1)$ b. $x^2 - 5x + 6$ $(x-3)(x-2)$ c. $x^2 - 4x + 4$ $(x-2)(x-2)$
   d. $x^2 - 5x + 4$ $(x-4)(x-1)$ e. $x^2 - 9x + 8$ $(x-8)(x-1)$ f. $x^2 - 6x + 8$ $(x-4)(x-2)$
   g. $x^2 - 13x + 12$ $(x-12)(x-1)$ h. $x^2 - 8x + 12$ $(x-6)(x-2)$ i. $x^2 - 7x + 12$ $(x-3)(x-4)$

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3. Factorise the following: $(x+16)(x-2)$

a. $x^2-4x-32$ $(x-8)(x+4)$
b. $x^2+14x-32$
c. x^2-x-6 $(x-3)(x+2)$
d. x^2+5x-6 $(x+6)(x-1)$
e. x^2-7x-8 $(x-8)(x+1)$
f. x^2+2x-8 $(x+4)(x-2)$
g. $x^2-5x-36$ $(x-9)(x+4)$
h. $x^2+16x-36$ $(x+18)(x-2)$
i. $x^2-35x-36$ $(x-36)(x+1)$

https://www.geogebra.org/m/JdjMFktK

Exercise 6

1. What makes these examples different? Factorise the following: Perfect squares

a. x^2+2x+1 $(x+1)^2$
b. x^2+6x+9 $(x+3)^2$
c. $x^2-8x+16$ $(x-4)^2$
d. $x^2-10x+25$ $(x-5)^2$
e. $x^2+12x+36$ $(x+6)^2$
f. $x^2-20x+100$ $(x-10)^2$
g. $x^2-18x+81$ $(x-9)^2$
h. $x^2+16x+64$ $(x+8)^2$
i. $x^2+26x+169$ $(x+13)^2$
j. $x^2-24x+144$ $(x-12)^2$

2. If $x^2-22x+m=(x-k)^2$, determine the values of m and k.
$x^2-22x+m = x^2-2xk+k^2$

$-22x = -2xk$
$22 = 2k$
$11 = k$

$m = k^2$
$m = 11^2$
$m = 121$

3. Show that $x^2-2x+1=(x-1)^2$ if the value of $x=-\frac{1}{2}$
LHS = RHS

Example 5: Using Substitution to Factorise: $(x+1)^2+21(x+1)-100$

let $(x+1)=k$

$\therefore k^2+21k-100$

$=(k+25)(k-4)$

$=(x+1+25)(x+1-4)$

$=\underline{(x+26)(x-3)}$

Exercise 7: Now try these:

a) $(x-1)^2-2(x-1)-63$ $(x-1-9)(x-1+7)$ $(x-10)(x+6)$
b) $(t+2)^2+22(t+2)+121$ $(t+2+11)(t+2+11)$ $(t+13)^2$
c) $(y-3)^2+20(y-3)+36$ $(y-3+18)(y-3+2)$ $(y+15)(y-1)$
d) $(b+1)^2-12(b+1)+36$ $((b+1)-6)^2$ $(b-5)^2$
e) $(x+2)^2+3(x+2)-54$ $(x+2+9)(x+2-6)$ $(x+11)(x-4)$
f) $(k+4)^2+(k+4)-30$ $(k+4+6)(k+4-5)$ $(k+10)(k-1)$

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TYPE 3: Trinomials: The coefficient of x^2 is not equal to 1 $(a \neq 1)$.

Make your own notes about each example.

Example 1: Factorise: $3x^2+10x+7$

$=(3x+7)(x+1)$

Example 2: Factorise $3x^2-16x-12$

$=(3x+2)(x-6)$

Example 3: Factorise $2x^2-9xy-5y^2$

$=(x-5y)(2x+y)$

Example 4: Factorise $-15p^2+23pq-6q^2$

$=-(15p^2-23pq+6q^2)$

$=-(3p-q)(5p-6q)$

Factorise Trinomials of the form: ax^2+bx+c

Exercise 8 *Factorise the following:*

a) $2x^2+x-3$ $(2x+3)(x-1)$

b) $2x^2-x-3$ $(2x-3)(x+1)$

c) $3x^2+7x+2$ $(3x+1)(x+2)$

d) $10x^2+3x-1$ $(5x-1)(2x+1)$

e) $3x^2+5x-2$ $(3x-1)(x+2)$

f) $8x^2-3+2x$ $8x^2+2x-3$ $(2x-1)(4x+3)$

g) $3x^2-19x+20$ $(3x-4)(x-5)$

h) $5x^2-8x-36$ $(5x-18)(x+2)$

i) $24x^2-2xy-y^2$ $(4x-y)(6x+y)$

j) $15a^2+13ab-28b^2$ $(a-b)(15a+28b)$

k) $5p^2+15q^2+28pq$ $5p^2+28pq+15q^2=(5p+3q)(p+5q)$

l) $8x^2-2xy-y^2$ $(2x-y)(4x+y)$

m) $-6a^2-11a-5$ $-(6a^2+11a+5)=-(6a+5)(a+1)$

n) $-2q^2+7q+4$ $=-(2q^2-7q-4)$ $=-(q-4)(2q+1)$

https://www.geogebra.org/m/kzz6n2a9#material/VqeZggze

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a) $(x-1-2)(2(x-1)-3)$
$(x-3)(2x-5)$

b)

2. Factorise fully: Hint-use substitution.

$2y^2 - 7y + 6 = (y-2)(2y-3)$

a) $2(x-1)^2 - 7(x-1) + 6$

b) $6(y+2)^2 - 11(y+2) - 10$

$6x^2 - 11x - 10 = (2x-5)(3x+2)$
$(2(y+2)-5)(3(y+2)+2)$
$(2y-1)(3y+8)$

c) $5A^2 + 14A + 5 = (5A+4)(A+2)$

c) $5(2x+1)^2 + 14(2x+1) + 8$

$(5(2x+1)+4)(2x+1+2)$
$(10x+9)(2x+3)$

d) $15(a-b)^2 - (a-b) - 2$

$15B^2 - B - 2 = (5B-2)(3B+1)$

$(5(a-b)-2)(3(a-b)+1)$
$(5a-5b-2)(3a-3b+1)$

TYPE 4 : Factorise by Grouping

When there are more than 3 terms in the expression, the terms are grouped together in order to use a common factor and/or difference between squares to factorise the expression.

Example 1: Factorise: $6bx - 15cx + 10cy - 4by$

$= 3x(2b-5c) + 2y(5c-2b)$
$= 3x(2b-5c) - 2y(2b-5c)$
$= (2b-5c)(3x-2y)$

Example 2: Factorise: $10x^5 - 15x^2 - 4x^3 + 6$

$= 5x^2(2x^3-3) - 2(2x^3-3)$
$= (2x^3-3)(5x^2-2)$

Example 3: Factorise: $x^2 - 2x - y^2 - 2y$

$= x^2 - y^2 - 2x - 2y$
$= (x+y)(x-y) - 2(x+y)$
$= (x+y)(x-y-2)$

Example 4: Factorise: $5a + b(a+5) + b^2$

$= 5a + ab + 5b + b^2$
$= a(5+b) + b(5+b)$
$= (5+b)(a+b)$

Exercise 9

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